

Bank- and Country-based Determinants of Banks Performance in Asia: A Corporate Governance Approach

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JEL classification: C23; G21; L2

Key works: Bank Performance, Net Interest Margin, Return on Assets, Asia, Panel Data,

Abstract:

Asian banking sector has encountered several financial distresses over the last few decades as well as rigorous reforms. The main objective of this paper is to analyze the impact of bank-based and institutional-based drivers on the Asian banks' performance. Hence, we examine the behavioral bank model using both net interest margin and return on assets. In order to deal with possible endogeneity, we use the two-step system estimator with adjusted standard errors through the Generalized Method of Moments (GMM). With a dataset of 2,911 bank observations for 11 Asian countries we find evidence that the ownership structure and the capital ratio are critical determinants of the banks' profitability which allow to control in a certain extent the agency problems. The revenue diversification impacts negatively on the net interest margin whilst the asset diversification is a source of value. Examining the institutional factors of banks performance, we find that the better the financial and the regulatory structures, the lower the capacity of banks to generate non-competitive abnormal profits. The ultimate affected from weak competition is the consumers who bear the burden with higher prices than competitive prices for availing banking services.

Key works: Banks' profitability, agency problems, corporate governance, Asian banking system, capital ratio, regulatory system

JEL Classification: G21, G38, E44

1. Introduction

From the global perspective, Asian economy has been gaining significant attention over the last decades (Peng, Bhagat, & Chang, 2010; Pilbeam, 2005). The Asian financial crisis in 1997 has questioned the financial stability in this region. On the other hand, the financial crisis of 2008 had no significant effects on Asian economy (Soedarmono, Machrouh, & Tarazi, 2013). Woo, Sachs, and Schwab (2000) reasoned unfettered bank competition as the main cause for the 1997 crisis. The aftermath of the 1997 crisis is an overall banking restructure and reform of corporate governance among these countries (Soedarmono, Machrouh, & Tarazi, 2011). Bank consolidation (e.g. mergers and acquisitions –M&A) and foreign direct investment (FDI) in Asian countries have scored significant upward changes after the 1997 crises (Moshirian, 2008). Moreover, as Soedarmono et al. (2011) argued, the status of corporate governance reforms has been made to overcome the imprudent strategies.

In this study, we select the Asian banking sector for the following reasons: firstly, over the last decades, Asian countries have faced dramatic government intervention in financial reforms and deregulations (Cook, 2008). Secondly, Asian countries have confronted a significant financial pressure during the Asian crisis of 1997 and then, surprisingly a huge shift in both financial restructuring and globalization effort in 21st century. Next, private sector of Asian economy is termed as vulnerable due to the presence of poor accounting standards, weak transparency in

management and political instability (Pontines, 2008). Such corporate vulnerability in the long-term may cause additional financial instability (Stiglitz & Weiss, 1981). Last but not least, Asian banking stability remains an existing debate to both the academics and professionals worldwide. Moreover, the existence of bank competition, previous crises and ongoing regulatory changes are also connected to this study.

The specific contribution of this paper is that it examines performance of Asian banks considering both the bank and country specific issues. Among the earliest studies of bank performance in Asian countries, Lee and Hsieh (2014) examined impact of foreign ownership on financial stability, but no consideration to corporate governance is analyzed. Similarly, Lee, Hsieh, and Yang (2014) investigated bank diversification considering financial restructuring among selected Asian countries, but once again, it fails short in building hypotheses upon governance issues. Other major studies have focused on individual Asian countries only such as Berger, Hasan, and Zhou (2010), which limits the likelihood of extrapolation of results to a wider context. Thus, this paper reveals a holistic view of major drivers of Asian bank performance and fills some of the existing gaps in the literature.

The initial research on bank performance was focused on the determinants of bank interest margins. The seminal paper of Ho and Saunders (1981) has been the theoretical framework for most of the further research on the drivers of bank net interest margins. The dealership model of Ho and Saunders indicates that the optimal net interest margin is a function of risk aversion, the size of bank concentration, the interest rate risk on deposits and loans, and the degree of market competition. This model has been widely used and extended/improved in the literature.¹

In the same way as Ben Naceur and Omran (2011), in this paper we follow an alternative approach focused on performance analysis using both net interest margins and return on assets with a more eclectic one-step estimation process based on a behavioral model of the banking firm. Previous literature focused on the study of bank profitability also considers this pragmatic approach where the determinants of profitability are classified as internal and external factors (Bourke, 1989; Demirgüç-Kunt & Huizinga, 1999; Goddard, Molyneux, & Wilson, 2004; Molyneux & Thornton, 1992; Saona, 2011; B. Short, 1979). Past literature usually considers that the internal drivers of bank profitability are management- controllable factors which account for the intra-firm differences in commercial bank profitability, given the external environment.² While the external factors are the set of those taken for granted, away from bank's control, and are expected to affect positively and/or negatively the banks' business (Athanasoglou, Brissimis, & Delis, 2008; Ramlall, 2009; Sufian & Habibullah, 2009). These variables are basically determined by the legal and institutional framework, the financial system, and the peculiarities of the economic (macroeconomic) setting where a bank performs its operations (Demirgüç-Kunt, Laeven, & Levine, 2004). Athanasoglou et al. (2008) investigated the determinants of bank profitability in a single equation framework composed with bank-specific (which involves operating efficiency, financial risk, and banks' size), industry-specific (which includes variables which are not the direct result of managerial

¹ For instance, Lerner (1981) discusses critically that certain assumptions behind the model might lead to errors. Afterwards, Allen (1988) extends the Ho and Saunders model to consider the case of loan heterogeneity. In the context of European banks, Carbó Valverde and Rodríguez Fernández (2007) use a multi-output framework to show that the relationship between bank margins and market power varies significantly across bank specializations. Focused on the European Union banks, Maudos and Fernández de Guevara (2004) widen the Ho and Saunders model to take banks' operating costs explicitly into account. Additionally, Saunders and Schumacher (2000) use a multicountry setting and decompose bank margins into a regulatory component, a market structure component and a risk premium component.

² They also break down this classification into both financial and non-financial (off-balance sheet) statement variables.

decisions) and macroeconomic-specific (cyclical output and expected inflation, for instance) determinants of profitability.

The general goal of this paper is to determine the effect of bank-based and institutional-based drivers of performance of Asian banks. Recently, Lin, Chung, Hsieh, and Wu (2012) addressed diversification activities in the banking industry that has become an important trend, however the existing literature on the determinants of bank interest margins does not address the effect of such diversification. After the Asian crisis in 1997, the domestic owners of the Asian banks failed to recapitalize the banks. Therefore, regulators allowed foreign banks to come and participate in the national banking system. Again, to protect the bank stability, banks in Asian region adopted international standards (e.g., Basel accord) which created an additional filter in providing credit to risky investors as well as caused extra requirements in capital reserve. These two reasons have triggered Asian banks, in particular, to go for revenue diversification (Lee et al., 2014). Therefore, the specific goal of this paper is, besides considering the internal and institutional determinants, to analyze how diversification of banks' business determines bank performance.

The remaining of the paper is structured as follows. Section 2 provides a description of the related literature and the research hypotheses. Section 3 develops the methodology applied in the empirical analysis and describes the variables used. The main results are shown in Section 4 and after this, in the final section, the paper draws the conclusions and policy implications.

2. Related literature and research hypotheses

2.1. The corporate governance argument as driver of the performance in the banking industry

In the wave of financial crises, corporate governance in the banks has received steadfast attention due to associated corporate scandals (Erkens, Hung, & Matos, 2012). To be precise, the scandals are only one of the underlying structural reasons for which corporate governance has become a critical success factor: the government regulations and interventions, worldwide trend of privatization of the banking industry, the increased market competition, and the integration of capital market (Becht, Bolton, & Röell, 2003). Good corporate governance is assumed to mitigate the adverse possible effects of agency risk (Min & Smyth, 2014). Prior literature has underlined a positive association of corporate governance with economic development. Claessens (2006) summarized the channels as a better option for external financing, lower cost of capital, higher firm value, optimal allocation of resources, and stakeholders' acceptability. Since banks are the most influential partner of countries' economic growth and development (Levine, 1997), a higher level of corporate governance practices is expected in the banking sector. Polo (2007) revealed that corporate governance of a bank is significantly different from the other nonfinancial firms because of their opaqueness and government regulation. Similarly, a recent study by Tan (2014) also found that corporate governance problems at banks are mostly for governance structure, government safety-nets, and opaqueness.

2.2. Measures of performance

Following Demirgüç-Kunt and Huizinga (1999) we use two measures of banks' performance. The first one is the bank interest spread or net interest margin. We use the ex-post spreads which are measured as the difference between the banks' actual interest revenues and their actual interest expenses. This ex-post spread differs from the ex-ante spread by the amount of loan defaults. The ex-post spread is a more useful measure because it controls for the fact that banks with high-yield, risky credits are likely to face more defaults.

The second measure of bank profitability is approached by the return on assets as opposed to the return on equity. Return on assets is a financial ratio used to measure the relationship of earnings to total assets. Jahan (2012) recently reported that the return on assets is the best and most widely used indicator of earnings and profitability supplemented by return on equity (ROE) and return on deposits (ROD). In fact, the return on assets assesses how efficiently a bank is managing its revenues and expenses, and also reflects the ability of the bank's management to generate the net income accruing to the bank from non-interest activities. Additionally, Demirgüç-Kunt and Huizinga (1999) argued that the problem in some developing countries is that banks operate with extremely low equity capital, often supported by implicit state guarantees, which inflates their return on equity. It can be observed that the alternative measures for the return on equity are substantially higher than those for the return on assets as argued by Demirgüç-Kunt and Huizinga (1999). Therefore, since this work is focused on Asian economies, we use the return on assets as a proxy for performance to minimize the measurement biases.

2.3. Drivers of banks' performance and their associated hypotheses

2.3.1. Internal determinants

Capital ratio: One of the major drivers of banks' performance is their capitalization. Even though there are not clear findings regarding the relationship between a bank's capital ratio and its performance, Berger (1995b) argued that there is a positive relationship between the capital ratio and the bank's profitability. Such a relationship is supported by two complementary arguments. The first one is addressed by the expected bankruptcy costs hypothesis; according to which banks increase their capital ratio whenever the exogenous factors increase the expected bankruptcy costs. Therefore, banks' capital ratio increases in order to reduce the likelihood of default and thereby lowering the expected value of bankruptcy costs. The second argument is based on the signaling hypothesis. According to this hypothesis a positive relationship between banks' performance and their capital ratios is expected. In this case, management might be willing to convey information to the market concerning its future prospects and capacity to generate profits. As a result, a signaling equilibrium might exist where banks expect to have improved future performance exhibiting higher capital ratios.

Additionally, a higher capital ratio involves with a higher flexibility to take advantage of new business opportunities, mostly when banks have financial constraints derived from unexpected losses in business operations (Goddard et al., 2004). From a sample of Middle East and North Africa countries, Maudos and Fernández de Guevara (2004) found that bank capitalization has a positive and significant impact on banks' net interest margin, cost efficiency, and profitability.

Besides these arguments, the literature also provides results for a negative relationship between banks' capital and banks' performance. The traditional view of bank profitability suggests that a higher capital-asset ratio is linked with a lower return on equity because a higher equity capital ratio decreases the risk on equity and the tax subsidy provided by interest deductibility (Berger, 1995a). An excessively high capital ratio might denote that a bank is operating cautiously or over conservatively, and ignoring potential profitable growth opportunities (Saona, 2011). This argument involves higher opportunity costs of capital when the capital ratio increases.

Berger and Bonaccorsi di Patti (2006) provided an argument for the negative relation between performance and capital. They used the efficiency-risk hypothesis where more efficient banks tend to choose relatively low capital ratios, as higher expected returns from the greater profit efficiency substitute equity capital to some degree, in terms of protecting the banks against financial distress, bankruptcy, or liquidation. An additional argument with supporting the negative relationship between capital ratio and performance in the banking industry comes from

the agency costs hypothesis (Jensen, 1986; Jensen & Meckling, 1976). According to this, high leverage –or, in other words, a low capital ratio– reduces the agency costs of outside equity and increases firm value by constraining or encouraging the managers to act more for the interest of shareholders. Higher leverage can mitigate conflicts between shareholders and managers concerning the choice of investment (Myers, 1977), the amount of undertaken risk (Jensen & Meckling, 1976), among others. Therefore, this higher control on managers would lead towards a better bank performance. Empirically, Berger and Udell (2006) found evidence consistent with the agency costs hypothesis which means a relatively low equity capital ratio in banking is associated with higher profit efficiency.

Most of the previous empirical literature studied the relationship between banks' performance and equity capital ratio using a monotonic linear relationship between these variables (Ben Naceur & Omran, 2011; Chaudhry, Chatrath, & Kamath, 1995; Goddard et al., 2004; Molyneux, Remolona, & Seth, 1998; Molyneux & Thornton, 1992). Saona (2011) however, for a sample of US banks, considered a non-monotonic relationship which seems to fit better the theoretical arguments. In the same vein, Baele, De Jonghe, and Vander Vennet (2007) found that there is a quadratic relationship between equity capital ratio and the banks' franchise value. Additionally, the findings of Memmel and Raupach (2010), using monthly regulatory data of large German banks, revealed that there exists a target level of capital structure for a substantial percentage of banks. Therefore, it might be hypothesized that there is a quadratic relationship between the equity capital ratio and the bank's performance which leads to observe an optimal capital structure maximizing the bank profitability.

Corporate ownership structure: Stating from Berle and Means (1932) on, the finance literature contains an extensive debate for ownership structure and firm performance. There are several dimensions of ownership structure that varies from the governance of a sole ownership to the widely diverse shareholders. Mixed results are found from the past studies. For instance, studying the incentive theory, Morck, Shleifer, and Vishny (1988) suggested a roof-shaped relationship between firm value and management ownership. This means that the firm value rises as the board ownership increases, but up to a certain point. After such point, the firm value declines as board ownership concentration increases. A similar result is assured by H. Short and Keasey (1999) and Perrini, Rossi, and Rovetta (2008). Therefore, at low levels of the managerial ownership, according to the agency theory, executive shareholdings encourage managers to improve firm value –convergence or alignment of interests hypothesis, since managers bear a proportion of wealth as shareholders (Demsetz & Villalonga, 2001; Jensen & Meckling, 1976). However, when the interests of managers and shareholders are not fully aligned, higher stock ownership can give managers much power to pursue their interest without fear of being fired or punished (Alves, 2012). This conflict is known as the entrenchment problem and takes place at high levels of managerial ownership, impacting negatively the performance of a bank. Therefore, we might expect a non-monotonic relationship between the managerial ownership and the bank performance supported by the interaction of the convergence and the entrenchment hypotheses.

In terms of the ownership concentration, basically we can distinguish a vertical and a horizontal agency conflict within a bank. The vertical agency problem (or type I agency conflict) arises when there are discrepancies in the interests between managers and shareholders (Shleifer & Vishny, 1986). In this particular case, a high ownership concentration leads to a convergence of interests between both groups and consequently the performance increases. Therefore, this known as the supervision hypothesis suggests that a firm's performance increases as the ownership concentration rises. However, when the corporate ownership structure is more concentrated than the optimal level in order to reduce the agency conflicts, it triggers the expropriation problem. This is known as the horizontal agency problem (or type II agency

conflict) which is caused when the majority shareholders use their decision making power in their own best interest; which not necessarily corresponds with the one held by minority shareholders (de Miguel, Pindado, & de la Torre, 2004, 2005; Saona, 2014). Therefore, at high levels of ownership concentration it might be expected that the wealth expropriation of minority shareholders by large (e.g. blockholders) shareholders erodes the performance of banks. This reasoning drives our second hypothesis concerning the corporate ownership structure which establishes that the conjunction of the supervision and expropriation hypotheses stresses an inverse U-shaped relationship between the ownership structure and the performance of banks.

Functional Diversification: Fee-based and financial advising banking services constitute nowadays an additional source of revenues in Asian banks. Studies on functional diversification provide mixed results (Trujillo-Ponce, 2013). Whilst Mercieca, Schaeck, and Wolfe (2007) for a sample of small EU banks found no effect of diversification on bank performance; Cybo-Ottone and Murgia (2000) also for a set of EU banks found a significant positive abnormal returns associated with product diversification into insurance companies. Wall and Eisenbeis (1984) found a negative correlation between bank earnings and securities broker/dealer earnings for a sample of US banks. DeYoung and Roland (2001) found that fee-based activities raise the overall level of volatility of earnings. Berger et al. (2010) conclude that all dimensions of diversification considered in their analysis (loan deposits, assets, and geography) were associated with higher costs and reduced profits.

Existing literature on revenue diversification and bank performance in the Asian region has exhibited a mixed result. M. Nguyen, Skully, and Perera (2012a) examined 153 banks from five ASEAN countries and revealed a non-linear relationship between banks' market power and diversification. With a lower level of market power, banks are found well diversified and vice-versa. Similar findings are observed by M. Nguyen, Skully, and Perera (2012b) for the South Asian context. For Taiwanese banks, Lien and Li (2013) found an inverse U-shaped effect on performance from the revenue diversifications of the examined banks. However, Lee et al. (2014) examined 29 Asia-Pacific countries and articulated that the effect of diversification differs across the countries based on the market power and structure of the banking sector. They also suggested that, unlike the developed western economies, the revenue diversification has no significance with a higher level of non-interest income capacity of the banks.

Despite all the previous arguments, in the Asian context, fee-based and financial advising based banking services constitute an additional source of revenues. As a process of liberalization and international integration in Asia, the traditional banking business of the 80s has been gradually pursued to functional diversification through activities such as electronic pursue, delivery channels, clearing systems, investment banking, security tradings, hedge funds, foreign exchange, assurance, and other financial services able to generate revenue in a variety of different ways, including interest, fees, transaction fees, and commissions (Valdez, 2007). Consequently, it is expected that a positive impact of such functional diversification on the banks' performance.

Bank size: An additional intra-bank driver of its performance comes from the relative size of the bank. The effect of banks' size on their performance underlies in an optimal size which maximizes the profitability. Athanasoglou et al. (2008) suggested that in general the effect of bank size on profitability has been proved to be positive to a certain extent. They also revealed that for banks which became extremely large, the effect of size could be negative due to bureaucratic reasons. On this respect, Goddard et al. (2004) introduced the economies of scale and showed that, at small size of assets, banks can take advantage of the economies of scale, but they become exhausted as the size of assets increases. These findings, therefore, suggest that economies as well as diseconomies of scale drive banks' performance. The model of

Maudos and Fernández de Guevara (2004), following Ho and Saunders (1981), predicted that the unit margins are an increasing function of the average size of operations. The justification is that, for a given value of credit risk and of market risk, an intermediation operation of greater size would mean a greater potential loss, so the bank will require a greater margin. Likewise, the potential loss will be greater for those banks in which the volume of credits granted is greater. The empirical literature has found that, for instance, banks with larger operations bear high risk, and, thus, charge higher margins, improving their performances (Maudos & Solís, 2009). In the opposite way, Hawtrey and Liang (2008) found that the size of bank transactions is inversely related to banks' spread. They argued that their finding is not surprising because transaction size is likely to be a proxy for scale of economies. In contrast to Maudos and Fernández de Guevara (2004), Hawtrey and Liang (2008) predicted margins to decline as bank scale increases, on account of the standard cost economies of scale effect. For a sample of different types of Chinese banks, Heffernana and Xiaoqing (2010) found that bank size does not influence performance. Therefore, banks' size-performance relationship might be positive or negative depending on the existence of economies or diseconomies of scale (Athanasoglou et al., 2008).

Credit risk: The credit risk might also drive the banks' performances. Theoretically, it is suggested that increased exposure to credit risk is usually associated with decreased firm profitability (Athanasoglou et al., 2008). Therefore, the way banks can improve their performance is by means of screening and monitoring their credit risk. Additionally, central banks set some specific standards for the level of loan-loss provisions to be adopted by the country's banking system; which means, credit risk is a predetermined variable. Empirically, using both cross-section and pooled time-series regressions for a sample of US banks, Miller and Noulas (1997) suggested that the more financial institutions are exposed to high-risk loans, means that the higher the accumulation of unpaid loans and the lower the profitability. Nevertheless, in their recent work for Asian countries, Lin et al. (2012) found that risky loans are positively associated with net interest margins, supported in the idea that banks with more risky loans will require a higher net interest margin to compensate for the greater risk of default. Therefore, we might hypothesize that the relationship between credit risk and banks' performance is positive for the Asian context where there is an active demand for larger margins in order to offset the additional credit risk of unpaid loans.

Bank loans: Bank loans are expected to be the main source of income and to have a positive impact on bank performance (Ben Naceur & Goaied, 2008). Assuming no change in other factors, if more deposits are converted into loans, then it is expected to observe a higher interest margin and profit. Empirical studies have found that higher loan ratios are associated with higher interest margins, suggesting that risk-averse shareholders seek for larger earnings to compensate higher credit risk (Ben Naceur & Goaied, 2008; Maudos & Fernández de Guevara, 2004). Nevertheless, Demirgüç-Kunt and Huizinga (1999) found that there is a negative correlation between bank loans and earnings before taxes, but when the bank loans are interacted with the GDP it becomes positive. This fact indicates that higher income level of bank's lending activities tend to be more profitable. Ben Naceur and Omran (2011) found that when market conditions enable the bank to provide additional loans with a profitable return/risk profile, this will, everything else remaining constant, improve the interest margin. Then, the expected empirical relationship between bank loans and performance should be positive.

Bank deposits: The demand for deposits represents the market profit opportunities (Berger, 1995a, 1995b; Berger & Bonaccorsi di Patti, 2006; Goddard et al., 2004). The demand for deposits is a primary source of agency problems due to the insurance protections given by the government (Berger, 1995a). In this case, one can expect a negative relationship between this variable and the bank's profitability. Nevertheless, the lack of competitive credit conditions in

Asia and the limited access to international capital markets lead banks to finance their growth with lower proportions of wholesale markets and higher proportions of customer deposits. Following the arguments of Trujillo-Ponce (2013), under this scenario, the deposits constitute a cheap and stable financial resource vis-à-vis with other financing alternatives, which supports a positive relationship between the banks' performance and customer deposits for the emerging markets like those in the Asian context.

2.3.2. *Institutional determinants*

The empirical literature has been quite prolific also in providing drivers of the performance of banks which are exogenously determined. This set of institutional-based determinants correspond to all those variables that are not defined internally by managerial decisions but by the current economic, financial system, and regulatory conditions. Demirgüç-Kunt et al. (2004), for instance, classify these kind of variables in regulatory variables, macroeconomic and financial system control variables, and institutional variables; while Ben Naceur and Omran (2011) classify them as regulatory impediments, macroeconomic variables, and financial and institutional development variables. Demirgüç-Kunt et al. (2004) is one of the first papers focused on the examination of the impact of these external determinants on bank interest margins, controlling for bank-specific factors and cross-country differences. Their findings are based on a sample of banks from more than 70 countries where bank-specific variables seem to explain a substantial part of the within-country variation in intermediary costs –banks performance, as well as the bank regulations. Moreover, the authors stress the fact that bank regulations cannot be viewed in isolation from the overall institutional framework. Therefore, we believe that including the institutional bank variables into the empirical analysis is a must, more than a suggestion. Thus, among our institutional-based variables we accounted for:

Bank concentration: according to both the market power and efficient-structure hypotheses, there is a positive relationship between the bank concentration and performance (Saona, 2011). Bourke (1989) and Molyneux and Thornton (1992) state that this positive correlation is due to increased market power yields monopolistic profits due to deviations from competitive markets. The collusion hypothesis also supports a positive relationship between banks' concentration and their performance. Demsetz (1973) finds that the assumption behind the collusion hypothesis is that the degree of market concentration exerts a direct influence on the degree of competition amongst the firms competing in a certain market. Then, highly concentrated markets will lower the cost of collusion and foster tacit and/or explicit collusion between firms. However, if the number of banks operating is large, the cost of collusion increases because it is more difficult to carry it out (Goddard et al., 2004). Nevertheless, if collusion is feasible, banks in the market will be able to earn monopoly rents. Then, we should expect a positive relationship between bank concentration and banks profitability.

Financial development: A wide-ranging literature is found on determinants of financial development that ensures banks performance (Arestis, Chortareas, & Desli, 2006; Guillaumont Jeanneney, Hua, & Liang, 2006; Hung, 2003). Levine, Loayza, and Beck (2000) examined South East Asian banks and concluded that a development in banking sector reflects a long-term economic growth and enhance financial development. More efficient and developed financial markets reduce bank profitability through higher competitiveness (Ben Naceur & Omran, 2011; Valdez, 2007). In the same line, Demirgüç-Kunt and Levine (2004) suggest that countries with developed stock markets might create a competitive environment that puts downward pressure on bank interest margins. Consequently, we should expect that as financial development improves, the bank interest margins should drop.

Legal enforcement and regulatory system: In banking literature, there is a major debate on the relationship between legal enforcement and bank performance. Examining the South East Asian banks, Levine et al. (2000) originated that the government restrictions on the foreign ownership were successful. The existing government restrictions provided an empty space to the local banks which eventually took advantage of it by increasing their lending and capturing larger market shares. Thus, government restrictions, in some cases, make the banking sector sustainable. Similarly, legal system that protects creditors resulted in better functioning investment and bank performance. La Porta, Lopez-De-Silanes, Shleifer, and Vishny (1998) examined 98 countries and identified that the countries with low legal framework to protect creditors have less performing banks in their economy. Levine (1998) examined bank performance in 43 countries and found a similar positive relation between bank performance and legal system to protect creditors. In the same way, in an empirical analysis for a large sample of countries, Demirgüç-Kunt et al. (2004) find that better legal enforcement and efficient regulatory systems are associated with lower levels of corruption, reducing the frictions in the financial system. However, Ben Naceur and Omran (2011), for a broad sample of Middle East and North Africa (MENA) countries, find that an improvement in the law and order decreases the cost of efficiency without affecting banks' performance. For the particular case of Asian banks, one should expect that the weak enforcement of the law and the high levels of corruption be ameliorated before more efficient legal and regulatory systems, which eventually would have a positive impact on banks' performance.

3. Methodology and variables' measurement

3.1. Methodology

Since we account with a data set compounded by individual banks observed during a number of years, the data base is suitable to be analyzed with the panel data approach. There are a number of benefits of using panel data. Arellano and Bover (1995) revealed that the use of panel data can estimate the dynamic effects that otherwise could not be possible in cross sectional or time series analysis. Himmelberg, Hubbard, and Palia (1999) stated that using panel data can help to overcome the constant and unobserved heterogeneity among the observations. Moreover, using panel data is suitable to attain a higher variability, less collinearity and higher efficiency.

Dealing with the endogeneity of banks' corporate governance is one the biggest challenges in recent empirical studies (T. Nguyen, Locke, & Reddy, 2014). The reach literature on corporate governance has found two main sources of such endogeneity problems –unobserved characteristics of corporations and simultaneity. On top of that, Wintoki, Linck, and Netter (2012) originated another source named dynamic endogeneity. This source is also supported by the previous theoretical studies, such as Hermalin and Weisbach (1998). In order to deal with these sources of endogeneity, we used the two-step system estimator with adjusted standard errors for potential heteroskedasticity as proposed by Blundell and Bond (1998). Originally, the Arellano and Bond (1991) estimator eliminates the individual fixed effects by transforming the regression in first difference and using GMM to estimate the parameters. The Arellano and Bover (1995) and Blundell and Bond (1998) estimator corresponds to an extension of the Arellano and Bond (1991) estimator, combining a system of regression in difference and still the ones proposed by Arellano and Bond (1991) in levels. In turn, the instrumental variables for the level regressions are composed by lags of the first-difference explanatory variables. Thus, under the assumption of stationarity of the panel, Blundell and Bond (1998) estimator is more efficient than the original Arellano and Bond (1991) estimator (Nakano & Nguyen, 2013; T. Nguyen et al., 2014; Saona & Vallelado, 2014).

Since consistency depends on the orthogonality of the instruments, Sargan overidentification test to check for exogeneity of the instruments will be used. Given that Sargan statistic is not robust to heteroskedasticity and autocorrelation, Hansen test will be used since it is not subjects to those drawbacks. The Hansen statistic follows a $\chi^2_{(q-k)}$ distribution, where q and k are the number of moment conditions and estimated parameters, respectively.

Regarding the autocorrelation, the test proposed by Arellano and Bond (1991) is applied to the first difference of the residuals, AR(1). Typically, the null hypothesis of no first-order correlation is not rejected. However, this is not considered a problem because $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{it-1}$ might be correlated with $\Delta\varepsilon_{it-1} = \varepsilon_{it-1} - \varepsilon_{it-2}$ given that both share the common term ε_{it-1} . Therefore, one must also perform the test for second order autocorrelation, AR(2). No rejection of the null hypothesis indicates that the moment conditions are valid.

3.2. Variables definition and model

The empirical analysis is carried out for a sample of 235 banks for 11 Asian countries (China, Indonesia, India, Japan, Korea, Sri Lanka, Malaysia, Philippine Islands, Singapore, Thailand and Taiwan), which compounds a set of 2,465 observations with an average of 10.5 observations per bank. The period of analysis goes from 2001 to 2013. The dataset is comprised with information from different sources. The banks' financial reports (e.g. balance sheet, income statement, and cash flow statements) and the market information (e.g. stock prices, market capitalization, and number of outstanding shares) as well as ownership structure information were obtained from Thomson Reuter's data set. Information on financial development at country level was obtained from the updated data based on the work of Beck, Demirgüç-Kunt, and Levine (2000) publicly available on the World Bank web page.³ Worldwide governance indicators at country level are based on the work of Kaufmann, Kraay, and Mastruzzi (2011) whose data set is also publicly available.⁴

Bank performance was measured by net interest margin (NIM) and alternatively by the return on assets (ROA). The net interest margin corresponds to the net interest income over the average earnings assets (Angbazo, 1997) and the return on assets is the pretax income over total assets (Goddard et al., 2004).

The capital ratio (CAP) was measured as the total common equity over total assets. Its quadratic form (CAP²) was also computed in order to test the non-linear relationship between this and the bank's profitability as suggested by our research hypothesis.

The corporate ownership structure will be measured by two alternative variables. For the insider ownership (INSOWN) we will use the ownership closely held which corresponds to the proportion of outstanding shares in hands of insiders. For the ownership concentration (OWN1) we use the percentage of shares in hands of the majority shareholder.

Three alternative measures of diversification were computed. The first revenue diversification index (DIVER1) was computed following Lee et al. (2014). This index takes the form $DIVER1 = 1 - \left[\left(\frac{NetIntInc}{OpInc} \right)^2 + \left(\frac{NetNonIntInc}{OpInc} \right)^2 + \left(\frac{LoanLossProv}{OpInc} \right)^2 \right]$. Where $OpInc$ denotes the total operating income and is equal to the sum of the absolute values of net interest income ($NetIntInc$), the net non-interest income ($NetNonIntInc$), and the provision for loan losses ($LoanLossProv$). This index ranges between zero –the bank is fully specialized in one source

³ The latest update took place in November 2013. Information can be downloaded from the permanent URL <http://go.worldbank.org/X23UD9QUX0>

⁴ The latest update took place in September 2014. Information can be download from www.govindicators.org

of income—, and 0.66 —the bank generates a fully balanced revenue mix from all three areas. Alternatively, we computed one index for asset diversity (DIVER2) and another one for revenue diversity (DIVER3) based on Laeven and Levine (2007), Baele et al. (2007), and Lin et al. (2012). The asset diversity index is based on stock variables, while revenue diversity is based on flow variables. These measures take the form of $DIVER2 \text{ or } DIVER3 = 1 - |2x - 1|$, where x is either the loans-to-assets ratio or the ratio of non-interest income to total operating income —the higher this ratio is, the more a bank relies on non-traditional bank activities. Consequently, diversity measured in this way (DIVER2 or DIVER3) takes values between 0 and 1 which means that the firm diversification increases with higher values.

The natural logarithmic transformation of banks' total assets (LNTA) is used as the proxy for the size of the bank (Athanasoglou et al., 2008; Fungáčová & Poghosyan, 2011; Maudos & Solís, 2009).

The credit risk was measured as the total loan loss provision over gross loans (CredRisk) (Athanasoglou et al., 2008). Bank loans are basically the proportion of total loans of total assets (LTA). The demand for deposit (DEPTA) was measured as the total deposit over total assets (Berger & Bonaccorsi di Patti, 2006).

Among the institutional-based variables we include in our estimations the bank concentration (BankConc) obtained from Beck et al. (2000) and measured as the market share of the three largest banks per the country. The financial development is measured by six proxies taken from the updated data base of Beck et al. (2000). These measures include: i) Deposit Money Bank Assets to GDP (DMBAGDP) which corresponds to the claims on domestic real nonfinancial sector by deposit money banks as a share of GDP, calculated using the following deflation

method: $\frac{0.5 \left[\frac{F_t}{P_{et}} + \frac{F_{t-1}}{P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$; where F is deposit money bank claims, P_e is end-of period Consumer

Price Index (CPI), and P_a is average annual CPI; ii) Other Financial Institution Assets to GDP (OFAGDP) computed as the claims on domestic real nonfinancial sector by other financial

institutions as a share of GDP, calculated using the following deflation method: $\frac{0.5 \left[\frac{F_t}{P_{et}} + \frac{F_{t-1}}{P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$;

where F is other financial institutions' claims, P_e is end-of period Consumer Price Index (CPI), and P_a is average annual CPI; iii) Private Credits by Deposit Money Banks and Other Financial Institutions to GDP (PCRDBOFGDP) which corresponds to the private credit by deposit money banks and other financial institutions to GDP, calculated using the following deflation method:

$\frac{0.5 \left[\frac{F_t}{P_{et}} + \frac{F_{t-1}}{P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$; where F is the credit to the private sector, P_e is end-of period Consumer Price

Index (CPI), and P_a is average annual CPI; iv) Stock Market Capitalization to GDP (STMKTCAP) which is the value of listed shares to GDP, calculated using the following

deflation method: $\frac{0.5 \left[\frac{F_t}{P_{et}} + \frac{F_{t-1}}{P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$; where F is the stock market capitalization, P_e is end-of period

Consumer Price Index (CPI), and P_a is average annual CPI; v) Stock Market Turnover Ratio (STTO) calculated as Ratio of the value of total shares traded to average real market

capitalization, the denominator is deflated using the following method: $\frac{\frac{T_t}{P_{at}}}{0.5 \left[\frac{M_t}{P_{at}} + \frac{M_{t-1}}{P_{et-1}} \right]}$; where T

is total value traded, M is stock market capitalization, P_e is end-of period Consumer Price Index (CPI), and P_a is average annual CPI; and iv) Private Bond Market Capitalization (PRBOND) which is the private domestic debt securities issued by financial institutions and corporations

as a share of GDP, calculated using the following deflation method: $\frac{0.5 \left[\frac{F_t}{P_{et}} + \frac{F_{t-1}}{P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$, where F is

the amount outstanding of private domestic debt securities, P_e is end-of period Consumer Price Index (CPI), and P_a is average annual CPI. The first three proxies of the financial development (DMBAGDP, OFAGDP, and PCRDBOFGDP) are used to measure the development of the banking system; whilst the other three proxies (STMKTCAP, STTO and PRBOND) are associated to the development of the capital markets.

Following Kaufmann et al. (2011), for the legal enforcement and regulatory system we include the following variables, resulting in a total of six dimensions of governance: (i) Voice and Accountability (VA) which is the process by which governments are selected, monitored, and replaced; (ii) Political Stability and Absence of Violence/Terrorism (PS) which measures the perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means, including politically-motivated violence and terrorism; (iii) the Government Effectiveness (GE) corresponds to the quality of public and civil services, and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government's commitment to such policies; (iv) Regulatory Quality (RQ) which measures the perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development; (v) Rule of Law (RL) which reflects the confidence of the agents in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence; and finally (vi) the Control of Corruption (CC) which measures the perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as "capture" of the state by elites and private interests.

In addition to these variables, we included dummy variables by country and year as control variables.

Berger and Bonaccorsi di Patti (2006) argue that bank profits show a tendency to persist over time, suggesting a dynamic model of the banking profitability. This persistence can be the result of the market competition barriers, banks' regulatory capital-ratios enforcement, informational opacity and/or sensitivity to external shocks, to the extent that there is a serial correlation between them (Memmel & Raupach, 2010). Then, these arguments suggest the application of a dynamic model of banking profitability which takes the following form:

$$NIM_{it} = \beta_0 + \beta_1 NIM_{it-1} + \sum_{j=1}^9 \delta_j X_{it} + \sum_{k=1}^2 \theta_k Y_t + \eta_i + \mu_t + \varepsilon_{it} \quad (1)$$

Where X_{it} represents the vector of j bank-based determinants of profitability of the i bank in the t period, Y_t is the vector of the k country-based determinants and η_i , μ_t and ε_{it} measure the individual effect, the temporal effect, and the stochastic error, respectively.

Specifically talking,

$$\begin{aligned} \sum_{j=1}^8 \delta_j X_{it} = & \delta_1 CAP_{it} + \delta_2 CAP_{it}^2 + \delta_3 OWN_{it}^2 + \delta_4 OWN_{it}^2 + \delta_5 LNTA_{it} + \\ & \delta_6 LTA_{it} + \delta_7 DEPTA_{it} + \delta_8 CREDRISK_{it} + \delta_9 DIVERSIF_{it} \end{aligned} \quad (2)$$

And

$$\sum_{k=1}^2 \theta_k Y_t = \theta_1 FINDEV_{it} + \theta_2 LAW_{it} \quad (3)$$

4. Results

4.1. Descriptive statistics

The interest margin of commercial banks is significantly related to a number of contextual variables (Demirgüç-Kunt & Huizinga, 1999). As it is seen in Exhibit 1, Indian and Sri-Lankan commercial banks have successfully secured higher rate of net interest margin over the years. Higher portion of banks asset as compared to total GDP, high inflation rate and stable economic conditions have combined effect on banks' *ex-post* spreads in case of these two countries (Peiris & Jayasinghe, 2015; Ray & Das, 2010). A sharp growth in NIM for Indian banks from 2000 to 2004 indicates the fact that some major entries of foreign banks into Indian market which eventually adjust in the adjacent years of global financial crisis. Overall, in the recent years a somewhat decreasing trend in NIM is observed for all the countries (figure 1). Nevertheless, the effect of normal business cycle is present in the NIM trends for all the countries (Adamauskas & Krusinskas, 2012; Albertazzi & Gambacorta, 2009). Table 1 displays the basic descriptive statistics for all variables and Table 2 shows their correlations coefficients.

4.2. Multivariate analysis

Table 3 displays the regressions between the independent variables determined at bank-level and the net interest margin (NIM). First of all, in all the regressions we observe that the independent variables are jointly statistically significant according to the Wald test. There is not second-order autocorrelation among the variables despite of in some regressions we do observe first-order autocorrelation. This fact does not invalidate our results. Since the very nature of the panel data, it is expected certain level of first-order autocorrelation at the standard 5% confidence level according to Arellano and Bond (1991). Regarding to the moment conditions, both the Sargan and Hansen over identification tests did not reject the over identifying restrictions, meaning that the set of instruments are orthogonal to the estimated residuals. Thus, the results reported in Table 3 are robust according to the standard diagnostic tests for the panel data. In addition to that, we used the Lind and Mehlum (2010) test to verify the non-lineal relationships.

4.2.1. Internal Determinants of Bank Profitability

First of all, in observe that the one-period lagged dependent variable (NIM_{t-1}) included in the right hand side of the equation shows systematically a positive and statistically significant coefficient. Besides its significance, the economic impact of such coefficient is relatively high. This means that the historical performance of the banks impact their current performance in a large extent. In other words, we observe that there is a large persistency in the profitability in the banking industry in Asian countries.

In all, except models 2 and 8 in Table (3), we can see that the CAP is significantly associated with banks' NIM. Unlike of most of the previous empirical literature (Ben Naceur & Omran, 2011; Chaudhry et al., 1995; Goddard et al., 2004; Molyneux et al., 1998; Molyneux & Thornton, 1992), we find that there is a non-linear relationship. As such, in the Asian context, bank's NIM increases as CAP increases but only up to 27.06% (computed as the average of the

critical value for CAP variable).⁵ Beyond this threshold in the capital ratio the net interest margin decreases. Unlike the traditional findings of positive relationship between bank's NIM and CAP (Berger, 1995b; M. Nguyen et al., 2012b), this quadratic relationship explains major characteristics of Asian banks. After the major three financial crises, the authorities of Asian countries restricted banks to be overleveraged. Consequently, the non-monotonic relationship starts with a positive impact of the CAP on the NIM and then shows a negative relationship between these variables. Now theoretically talking, both the expected bankruptcy costs as well as the signaling hypothesis support the positive relationship; while the negative relationship is explained by the opportunity cost of the equity capital since high capital ratios do not allow to take advantage of external sources of funds. Therefore, this finding allows accepting the empirical hypothesis of the existence of a quadratic relationship between the equity capital ratio and the bank's performance, which allows to observe an optimal capital structure which maximizes the profitability of banks. Such maximum value of the NIM ranges from a level of 13.64% (model 6 in Table 3) to 45.92% (model 1 in Table 3). According with the Lind and Mehlum (2010) test reported at the bottom of the table, the inverse U-shaped relationship between CAP and NIM is statistically significant, and therefore the null hypothesis of a monotone or U-shape is rejected for all the models in the table.

In models 3 through 10 we introduced the percentage of shares in hands of major shareholder and in hands of executives (INSOWN) as regressor against NIM. We observe that there is a quadratic relationship of INSOWN with NIM. More precisely, banks' NIM increases as INSOWN increases but up to 54.57% (computed as the arithmetic mean among the critical values of INSOWN in Table 3).⁶ So, when the ownership in hands of the major shareholders and in hands of the executives increases but not more than the threshold points, the convergence hypothesis of agency theory holds and in result the NIM increases. However, after the stated ratios, the type-II agency conflict may arise in case of Asian banks resulting in lower NIM. In this particular case, the entrenchment and the expropriation hypotheses of agency costs seem to erode the banks' performance by wealth expropriation of minority shareholders by large shareholders.

Our results support the hypothesis of a negative impact of the banks' size (LnTA) on their profitability. Athanasoglou et al. (2008) suggest that when banks are extremely large, the effect of the size is negative on the performance basically due a bureaucratic reasons. This argument seems to fit with the reality of Asian banks. In fact, the bank concentration measured by the market share of the three largest banks per country is 47.00%. This simple indicator describes a snapshot of how concentrated in only few large banks the banking system is in our sample. An excessive dimension generates inefficiencies, bureaucratic costs, overlapping and duplication of functions, among other costs (Goddard et al., 2004). Thus, our finding concerning the bank's size-performance relationship suggests that margins decline as bank scale increases, on account of the standard cost economies of scale effect, in the same line as the arguments provided by Hawtrey and Liang (2008) for a sample of OECD countries.

We do not have conclusive results concerning the total loans offered. Although the suggested hypothesis established a positive relationship between the loans (LTA) and the net interest margin (NIM), the results provided in Table 3 are conflicting because two models (8 and 9) support such hypothesis; but other three models (5, 6 and 10) show a negative and statistically

⁵ The computation of the critical value in the first regression of Table 3 is done by calculating the first derivative of this regression with respect to the CAP variable, and then by making this equal to zero. After that we have to solve for CAP which represents the point at which the net interest margin (NIM) is maximized. Specifically talking, this solution takes the form: $\partial \text{NIM} / \partial \text{CAP} = 0.0315 + 2(-0.0343 \times \text{CAP}) = 0$. Consequently, when CAP=45.92% the NIM is maximized. Idem calculations are done for all the other regressions which include CAP².

⁶ The critical values of INSOWN were calculated similarly to those for CAP variable in the footnote 5.

significant relationship. Consequently, we cannot accept our research hypothesis about the total loans. The deposits (DEPTA) however show consistently a negative impact on the NIM, *ceteris paribus*. Although we hypothesized a positive relationship between these variables supported on the argument that deposits might constitute a cheap and stable financial source in immature financial markets such those in Asia; the findings emphasize the fact that the larger demand for deposits implies more yields going towards the customers pockets. In other words, the demand for deposits represents the cost of debt; and when it increases the performance of the bank (NIM) decreases.

Most of the empirical literature on banking for developed countries encounters that there is a negative relationship between the credit risk (CreRisk) and the net interest margin as a result of lower performance caused by unpaid loans. Unlikely of these previous findings, our results for the Asian context indicate a positive relationship between the credit risk and the net interest margin at the standard 5% confidence level. This finding indicates that banks with more risky loans will require a higher net interest margin as a compensation for the greater default risk. These results are in line with those of Lin et al. (2012). What we can deduce out of it is that under immature banking systems such as those in emerging Asian countries, the banks transfer part of the insolvency costs to the clients by setting higher spreads between the borrowing and lending rates.

We have used three alternative measures of the business diversification of the banking industry. Divers1 and Divers3 measure the revenue diversification; whilst Divers2 is a proxy for the asset diversification. The consistency of our results concerning these variables might be seen in the last three regressions of Table 3. The variable Divers1 considers bank's revenue diversification activities between both non-interest income activities and interest income activities. The results in the eighth model in Table 3 show a negative and statistically significant relationship with the net interest margin for this variable. This fact suggests that non-traditional banking activities raise the overall operating costs and increase the volatility of earnings by impacting negatively on the bank performance. This finding is similar to the one of DeYoung and Roland (2001) for US banks. Additionally, Lepetit, Nys, Rous, and Tarazi (2008) have argued that banks expanding their business toward non-interest income activities –e.g. with higher diversification– show higher insolvency risk than less diversified banks. Consequently, this higher risk of insolvency presses downwards the performance of the banks as shown in our results.

Alternatively, we used an additional measure of the revenue diversification (Divers3) which takes into consideration the non-interest income over the operating income. In the same line than Divers1 variable, Divers3 is used as a proxy to measure in what extent the banks rely on non-traditional activities. In the last model of Table 3 it is observed that the higher the diversification in non-traditional banking activities, the lower the net interest margin. Therefore, the results are quite robust about the revenue diversification of the financial institutions in the Asian region, indicating that, on the one hand, there is an operating inefficiency of holding non-traditional activities; and on the other hand, there is a higher risk associated to more volatile profits which eventually erodes the performance of the banks.

However, concerning the asset diversification (Divers2) the results show that the net interest margin increases as long as bank assets are more diversified in activities different than the loans. Briefly, the results indicated that although the revenue diversification impacts negatively on the net interest margin; the asset diversification is a source of value which impacts positively the spreads between the borrowing and lending rates.

4.2.2. *External Determinants of Bank Profitability*

In this part of the analysis we entered into the general regression model the institutional variables which are those determined by the financial system and the legal and regulatory systems basically. The results in Table 4 concerning the bank-based variables might be considered as a robustness analysis. As it can be observed, all the major relationships described in section 4.2.1 are hold in Table 4. Briefly, the inverse U-shaped relationships for the capital-asset ratio (CAP) as well as the closely held ownership –inside ownership– (INSOWN) are still observed. Additionally, the banks size (LnTA), bank deposits (DEPTA), and the revenue diversification (DIVERS3) impact negatively on the capacity of banks to generate profits. Finally, as the unpaid loans increase (CREDRISK) the cost of the financial intermediation increases as well. Therefore, the customers are burdened with higher prices for financial services when the credit risk increases.

Concerning the institutional variables, they were classified in two big groups: i) the development of the financial system and ii) the development of the legal and regulatory system. The development of the financial system was measured through the development of the banking system (DBAGDP, OFAGDP, and PCRDBOFGDP) and through the development of the capital markets (STMKTCAP, STTO, and PRBOND); whilst the development of the legal and regulatory system was measured through six alternative governance indicators (VA, PS, GE, RQ, RL, and CC).

As we can observe in the first regression, the bank concentration (BANKCONC) holds the common view for emerging markets showing a negative and statistically significant impact on the banks' profitability. This indicates that monopolistic power creates an environment in which a few powerful banks stymie competition with deleterious implications for efficiency (Demirgüç-Kunt et al., 2004). Consequently, high concentration of the banking system is a signal of an uncompetitive and hence inefficient market which presses down the profitability (NIM) of its banks. The process of international integration of the Asian banking systems has been characterized by hostile takeovers and mergers with foreign financial institutions which have driven a higher concentration and market power in fewer banks (Mohd, 2014). This scenario has reduced the capacity of banks to generate profits efficiently impacting on their net interest margins negatively.

From regressions (2) through (7) the impact of the development of the financial system on the banks profitability is studied. We observe that the enhancement of the banking system as well as the improvements in the capital markets –the stock market and the market for corporate bonds– both impact negatively on the net interest margin (NIM). These relationships are different than the expected once. The financial systems in the Asian economies are still in an immature stage compared with other industrialized economies. In general, large asymmetries of information and market imperfections are still observed in these economies. All these facts describe a scenario where banks have taken advantage of in order to realize abnormal (or non-competitive) profits. Consequently, it seems to be that when the financial system achieves higher stages of development, such capacity to generate abnormal profits is minimized. In other words, when the financial markets are more efficient, dynamics, liquid, there is less room for monopolistic profits.

Concerning the legal enforcement and regulatory systems (see regressions (8) through (13) in Table 4), the findings are similar to Demirgüç-Kunt et al. (2004). We observe that the higher the enforcement of the law and of the regulatory system in general, banks have lower net interest margins. This indicates necessarily a need for better enforcement of the law and regulatory systems. When the legislation which protects the interests of investors is weak, the country is plagued with corporate scandals and corruption, and when the institutions lose their credibility as regulators, the financial institutions might generate abnormal profits by increasing their

spreads. This seems to be the measure adopted by banks to protect themselves against weak regulatory systems.

4.2.3. *Principal Component Factoring Analysis*

Since we account with a number of variables used as measures for the external determinants of the bank profitability and due all these variables are highly correlated (see Table 2), we cannot include them altogether in the same regression. Consequently, we opted by entering these variables individually as displayed in Table 4. In order to address this drawback in modeling the banks' profitability, we decided to apply the principal component factoring technique to enter all this variables in the same regression and take advantage of their informative content. The major benefits of this technique are that the factors created are not correlated, on the one hand, and the factors record a large extent of the variability of the individuals variables used in the estimation of the factors. Table 5 displays the number of factors generated for the variables used to measure the financial development and the variables used for the legal and regulatory system. In its Panel A we can observe that there is one factor whose Eigen value is higher than the unit as the standard discrimination value. This factor records about 63.20% of the variability of the six alternative variables used to measure the financial development by country (DMBAGDP, OFAGDP, PCRDBOFGDP, STMKCAP, STTO, and PRBOND). Panel B shows also that one factor is enough to record 87.40% of the variability of the variables used to measure the legal and regulatory systems (VA, PS, GE, RQ, RL, and CC).

These two factors (FinDevFact and LegalSysFact) have been entered into the regression analysis shown in Table 6. The three regressions in this table show that the factors are statistically significant. Consequently we can suggest that in a large extent the financial development, as well as the enforcement of the law channeled by the legal and institutional systems, is determinant of the performance of the financial intermediation industry in the Asian context. These results might also be seen as a robustness check of our institutional-based variables. Briefly, the better the financial system and the better the regulatory structures, the lower the capacity of banks to generate non-competitive abnormal profits. The ultimate affected by such weak competition is the consumer who is burden with higher prices than competitive prices for the banking services.

4.2.4. *Comparative Analysis Based on Differences in the Regulatory Systems*

We use Table 7 to compare the impact of the determinants of the NIM based on the cross-country differences in the regulatory systems. For doing so, we split the country sample in two groups according to the average of the legal and regulatory system variables as per country (average between VA, PS, GE, RQ, RL, and CC as showed in Table 7, Panel A). The group of countries with relatively high standards of corporate governance indicators is compounded by those countries with positive average indexes (e.g. Japan, Korea, Malaysia, Singapore, and Taiwan) whilst the group of relatively low indicators is formed by countries with negative average indexes (e.g. China, Indonesia, India, Sri Lanka, Philippines, and Thailand).

The regressions' outputs are shown in Table 7, Panel B. The findings are quite interesting and shed some light between the differences of these two groups of countries. First of all, the quadratic relationship observed in CAP for the whole sample so far, turns out differently when the country sample is split into these two groups. For those countries where there are relatively higher standards in the legal and regulatory systems, the banks increase their profitability when they use more equity capital. This relationship is supported by the expected bankruptcy costs hypothesis which suggests that banks increase their capital ratio to reduce the likelihood of

bankruptcy risk. However, for those countries with relatively lower standards of corporate governance in their legal systems, the inverse U-shaped relationship is observed between the capital ratio and the net interest margins. In this particular case, banks take advantage of external debt up to the point where the bankruptcy risk overwhelms the benefits of the leverage. The hypothesis of the inverse U-shaped relationship between CAP and NIM for the group of countries with relatively high legal and regulatory standards is trivially rejected.

Concerning the INSOWN variables we also observe appealing results. The first and third regressions show that as managers and the majority shareholder accrue more ownership, the agency problems between the principal and the agents are minimized and the bank profitability increases. However, when such concentration of ownership (in-between 43.32% and 48.19% as seen in the critical values of INSOWN) is higher than the one needed for effective monitoring both, the expropriation and the managerial entrenchment agency problems arise and the bank performance decreases. Consequently, when the country has relatively high legal and regulatory standards the composition of the ownership structure behaves somehow as an efficient corporate governance system. Therefore, we can conclude that under a scenario like this the regulatory system and the banks' ownership structure might become complementary corporate governance tools.

Nevertheless, the situation seems to be the opposite for countries with weak legal and regulatory standards (e.g. China, Indonesia, India, Sri Lanka, Philippines, and Thailand). In this case, only regression 4 enters coefficients for INSOWN and $INSOWN^2$ variables significantly. There we can observe that increases in the INSOWN impact negatively on the banks performance. This is a sign of a systematically high risk of expropriation of minority shareholders as well as entrenchment of managers. Therefore, we observe that the comparison between the two contexts of relatively high and low standards of corporate governance turn out to different behavior of the ownership structure across countries. For the set of countries with relatively weak legal and regulatory standards, the insiders' ownership does not work as a governance device. In fact, it encourages the opportunistic behavior of managers and the majority shareholders which eventually harms the net interest margin.

The findings show differential results concerning the size of the banks. Although for most of our previous results we observed a negative impact of the LnTA variable on NIM, in this case it seems to be that for the countries with relatively better legal and regulatory systems, banks take advantage of the economies of scale when their physical dimension increases. We do observe however that in the countries with low standards of corporate governance, banks are operating on the proportion of diseconomies of scale in their cost structure.

Although for the whole sample we could not accept nor reject the hypothesis about the loans (LTA). Once we control specifically for the efficiency of the legal and regulatory systems as governance tools, we observe that loans show a positive impact on the net interest margins. After controlling for this fact, this finding allows us to accept our hypothesis that once more deposits are converted into loans the profits and the net interest margin increase in the Asian context.

The rest of the results concerning the deposits, credit risk and the business diversification remain unaltered. The factors which measure the development of the financial system (FinDevFact1) and the legal and regulatory systems (LegalSysFact1) still show negative and statistically significant signs. This result provides even more robustness checks to our general findings that more developed financial systems reduces the likelihood of abnormal profits, on the one hand, and more efficient legal systems reduce the opportunistic behavior of making extraordinary benefits at expense of the customers, on the other hand.

4.2.5. *Robustness checks*

For robustness purposes we estimated the results of Table 6 throughout the alternative dependent variable measured as the return on assets (ROA). Basically, Table 8 replicates the results obtained in Table 6 but with this new variable. As it can be observed, the results are quite consistent with those obtained previously although some findings deserve to be highlighted. First of all, the return on assets as a measure of the banking performance seems to be more sensitive to changes to the capital-asset ratio than the net interest margin. The inverse relationship between the capital ratio and the ROA is also observed, but the firm seems to operate too conservatively at relatively lower levels of capital ratio (e.g. 23.93% computed as the average among the critical values of CAP in the three regressions Table 8). This finding makes sense because when excessive capital is used the banks do not take advantage of the financial leverage for creating further profits.

The ownership structure measured by the insiders voting rights (INSOWN) impacts the ROA roughly in the same way as the NIM does. In this particular case we observe that the asymmetric impact of the INSOWN variable on profitability reveals the sensitive effect of the ownership structure on the banks' profitability. In general, the profitability is increased as long as the stake in hands of the insiders does not exceed the 46.03% (computed as the average of the critical values of INSOWN in Table 8). After this point, the extraction of private benefits by the majority shareholders and managers at expenses of minority shareholders turns out to press down the general performance of banks in Asia.

The variable we used to measure the banks' size (LnTA) and the credit risk (CredRisk) change their signs when ROA enters the model as the dependent variable in comparison with those findings displayed in Table 6. On the one hand, when the bank's size increases, we observe that the profitability also increases. So, although previously we mentioned that larger banks seem to be operating under diseconomies of scale which influences negatively on their spreads; concerning their capacity in using the total assets to generate net income, the findings demonstrate that banks in Asia are relatively efficient on this respect. Higher levels of total assets push banks to generate more net income but cutting the net interest margin. What explains this somehow paradigmatic finding is basically the asset diversification. As it was observed previously, the asset diversification was found as a source of value, and now we can deduce how positive it is in generating net income for the bank.

The credit risk in the Asian banking industry seems to constraint the banks' capacity in producing after-tax benefits. The external determinants of banks profitability measured through the composite factors of financial development (FinDevFact1) and development of the regulatory and legal systems (LegalSysFact1) remain unchanged relative to those previously described.

As a summary, we can observe that the general findings discussed in the previous sections are quite robust independently on the variable used to measure the bank's profitability.

5. **Conclusions**

The trend of examining Asian economy has been upward for the last few decades. Researchers have been motivated by the dynamics of Asian financial sector, in particular, for a number of reasons: differences in responses to the recent financial crises, reforms of corporate governance, banking restructuring, and significant movement of foreign direct investment, among others. Moreover, the Asian banking stability remains an existing debate to the researchers for the existence of disturbed bank competition, poor accounting standards, weak transparency and ongoing regulatory changes. Unlike the traditional research of Asian bank performance, this

paper examines the impact of bank-based and institutional-based drivers of performance of Asian banks. This paper adopts the alternative approach of Ben Naceur and Omran (2011) for bank performance analysis considering both net interest margins and return on assets.

The results reveal that a number of contextual variables determine the bank's performance as well as variables determined by the banks themselves. First of all, we found a large persistency in Asian banks performance explained by the historical performance. Unlike previous results, we found a non-linear relationship between banks' profitability and the capital ratio. As such, in the Asian context, bank's profitability increases as the capital-assets ratio increases but only up to about 27%. Beyond this threshold, the banks' performance decreases. This quadratic relationship explains some characteristics of Asian banking; for instance, the expected bankruptcy costs and the signaling hypothesis support for a positive relationship; while the opportunity cost explains the negative relationship. After that, our models have examined the impact of percentage of shares in hands of major shareholder and in hands of executives. We observe that there is a quadratic relationship between these variables which indicates that the agency problems are solved in some extent. However, when the ownership structure is excessively concentrated, both the expropriation and the entrenchment risk augment. We do not have any conclusive result explaining impact of total loan offered on a bank's performance. In nutshell, the results of internal determinants of bank performance reveal that internal corporate governance mechanisms are still weak in Asian banking and this fact causes on the one hand, a detriment in value for the minority shareholders, and on the other hand, higher prices paid by customers for banking services.

Concerning the institutional variables the results reveal that the higher the enforcement of the law and of the regulatory system in general, banks have lower net interest margins, similarly to the effect of the development of the financial system. This indicates necessarily a need for better enforcement of the law and regulatory systems. When the legislation which protects the interests of investors is weak, the countries are plagued with corporate scandals and corruption, and when the institutions lose their credibility as regulators, the financial institutions might generate abnormal profits by increasing their spreads.

Policy recommendations might be derived from these conclusions. First, better legislations aimed to improve the internal and external corporate governance systems need to be developed. The Asian banking systems are operating with excessively concentrated ownership structures which allow managers and controlling shareholders to expropriate minority shareholders. Additionally, it is observed abnormal banks' profits which burden the pockets of customers. Higher enforcement of the law as well as codes on good corporate governance practices might help to mitigate these problems. Second, additional measures must be taken to enhance market competition in the Asian banking industry in order to reduce the abnormal profit. The results are quite consistent across different estimations and through alternative measures of banks' profitability.

Exhibit 1: Measures of Bank Performance by Year and Country

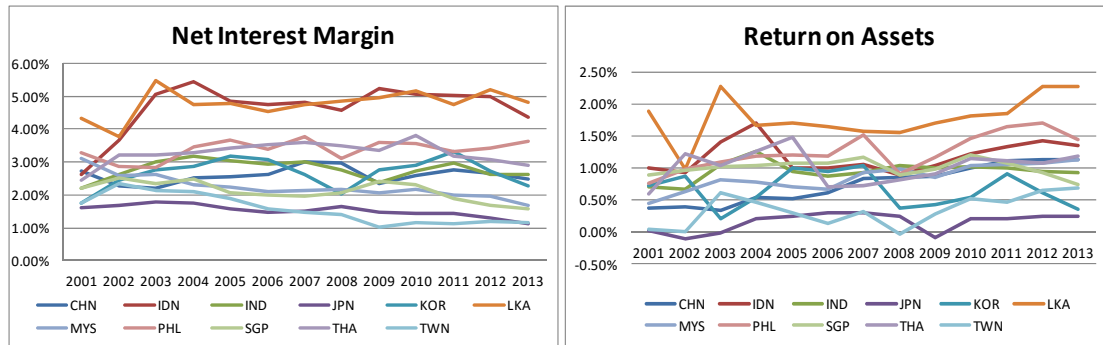
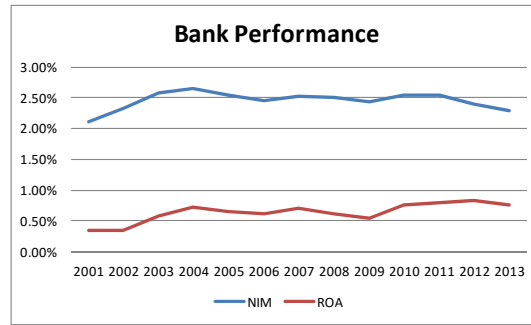


Table 1. Descriptive Statistics

Variable	Definition	Mean	Std. Dev.	Min	Max
NIM	Profitability	0.0245	0.0135	0.0047	0.0848
ROA	Profitability	0.0065	0.0065	-0.0045	0.0226
CAP	Capitalization Ratio	0.1717	0.0458	0.0150	0.5140
Own1	Ownership structure	0.2465	0.2542	0.0000	0.9996
INSOWN	Ownership structure	0.3311	0.2376	0.0030	0.9445
LnTA	Bank size	16.6611	1.7024	10.2920	21.8610
LTA	Loan	0.6546	0.1194	0.0000	0.9790
Depta	Deposits	0.7993	0.1443	0.0120	0.9630
CredRisk	Credit Risk	0.0067	0.0075	-0.0050	0.0410
Divers1	Diversification	0.4961	0.0957	0.0265	0.6667
Divers2	Diversification	0.6577	0.1884	0.0000	0.9996
Divers3	Diversification	0.4734	0.2467	0.0031	0.9993
BankConc	Bank Concentration	46.9127	15.4974	0.0000	100.0000
DBAGDP	Deposit Money Bank Assets to GDP	113.2857	52.6256	29.3071	195.4114
OFAGDP	Other Financial Institution Assets to GDP	101.6730	127.2544	0.0000	289.7113
PCRDBOFGDP	Private Credit by Deposit Money Bank and Other Fin. Inst. to GDP	110.4602	62.1713	17.1920	205.4089
STMKTCAP	Stock Market Capitalization	72.4028	33.2340	7.2719	242.7761
STTO	Stock Market Turnover Ratio	96.4757	48.4876	7.7361	377.2419
PRBOND	Private Bond Market Capitalization	23.7038	19.2347	0.0000	66.8345
VA	Voice and Accountability	0.3936	0.7020	-1.6820	1.1000
PS	Political Stability	0.0233	1.0248	-2.1180	1.3410
GE	Government Effectiveness	0.7527	0.7449	-0.4510	2.4300
RQ	Regulatory Quality	0.4962	0.6918	-0.7810	2.0290
RL	Rule of Law	0.5483	0.7835	-0.9660	1.7720
CC	Control of Corruption	0.4304	0.9280	-1.1340	2.4170
FinDevFact	Factor for Financial Development	-0.0068	1.0001	-1.6479	1.2087
LegalSysFact	Factor for Legal System	-0.0090	1.0023	-1.8104	1.5439
Obs.		2,465			

Table 2. Correlation Matrix

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
1 NIM	1.000																									
2 ROA	0.690	1.000																								
3 CAP	0.447	0.625	1.000																							
4 OWN1	0.405	0.376	0.195	1.000																						
5 INSOWN	0.450	0.397	0.278	0.673	1.000																					
6 LNTA	-0.373	-0.204	-0.405	-0.096	-0.220	1.000																				
7 LTA	0.173	-0.111	-0.169	0.082	0.072	-0.122	1.000																			
8 DEPTA	-0.265	-0.461	-0.576	-0.125	-0.134	-0.053	0.218	1.000																		
9 CREDRISK	0.458	0.135	0.147	0.260	0.274	-0.152	0.015	-0.185	1.000																	
10 DIVERS1	-0.003	-0.487	-0.225	-0.033	0.002	-0.160	0.275	0.313	0.334	1.000																
11 DIVERS2	-0.146	-0.026	-0.126	-0.080	-0.116	0.243	-0.707	0.013	-0.048	-0.129	1.000															
12 DIVERS3	-0.140	0.084	0.009	-0.097	-0.124	0.129	-0.104	-0.111	-0.026	-0.207	0.175	1.000														
13 BANKCONC	-0.061	0.080	0.249	-0.089	-0.010	-0.014	0.076	-0.182	-0.093	-0.197	-0.120	0.017	1.000													
14 DBAGDP	-0.740	-0.687	-0.458	-0.536	-0.529	0.427	0.028	0.334	-0.437	0.108	-0.005	-0.041	0.124	1.000												
15 OFAGDP	-0.514	-0.613	-0.416	-0.518	-0.447	0.200	-0.002	0.457	-0.362	0.220	0.015	-0.086	-0.128	0.837	1.000											
16 PCRDBOFGDP	-0.699	-0.681	-0.450	-0.545	-0.521	0.404	0.065	0.329	-0.420	0.137	-0.048	-0.058	0.106	0.991	0.866	1.000										
17 STMKTCAP	-0.451	-0.204	-0.111	-0.077	-0.188	0.276	0.040	-0.041	-0.166	-0.198	0.000	0.169	0.446	0.316	-0.056	0.254	1.000									
18 STTO	-0.471	-0.436	-0.303	-0.212	-0.309	0.434	0.047	0.039	-0.198	0.138	0.001	-0.124	-0.050	0.506	0.206	0.498	0.193	1.000								
19 PRBOND	-0.565	-0.531	-0.362	-0.397	-0.397	0.362	0.115	0.182	-0.346	0.016	-0.099	0.040	0.140	0.786	0.605	0.785	0.412	0.388	1.000							
20 VA	-0.501	-0.569	-0.294	-0.414	-0.383	-0.052	0.008	0.382	-0.257	0.224	-0.019	0.031	-0.312	0.517	0.662	0.513	-0.016	0.224	0.388	1.000						
21 PS	-0.657	-0.623	-0.295	-0.484	-0.449	0.270	0.085	0.287	-0.439	0.081	-0.077	-0.076	0.320	0.855	0.694	0.841	0.363	0.495	0.733	0.607	1.000					
22 GE	-0.701	-0.625	-0.284	-0.528	-0.493	0.245	0.065	0.251	-0.410	0.053	-0.072	0.015	0.365	0.872	0.693	0.852	0.491	0.403	0.768	0.602	0.936	1.000				
23 RQ	-0.661	-0.599	-0.226	-0.530	-0.472	0.197	0.101	0.205	-0.393	0.071	-0.129	-0.010	0.401	0.827	0.645	0.818	0.430	0.446	0.700	0.604	0.936	0.977	1.000			
24 RL	-0.731	-0.668	-0.340	-0.538	-0.528	0.237	0.057	0.292	-0.437	0.090	-0.038	0.012	0.204	0.876	0.739	0.853	0.425	0.448	0.735	0.725	0.924	0.967	0.942	1.000		
25 CC	-0.676	-0.637	-0.312	-0.542	-0.507	0.227	0.037	0.321	-0.447	0.093	-0.028	-0.040	0.259	0.894	0.808	0.878	0.334	0.377	0.690	0.673	0.932	0.961	0.940	0.969	1.000	
26 FinDevFact	-0.725	-0.693	-0.464	-0.522	-0.523	0.432	0.061	0.314	-0.425	0.117	-0.033	-0.039	0.097	0.982	0.821	0.981	0.348	0.563	0.862	0.530	0.865	0.877	0.832	0.882	0.878	1.000
27 LegalSysFact	-0.702	-0.661	-0.310	-0.542	-0.506	0.209	0.064	0.304	-0.428	0.103	-0.065	-0.013	0.240	0.870	0.754	0.854	0.374	0.430	0.722	0.733	0.956	0.977	0.968	0.987	0.980	0.873

Table 3. Regression Analysis for Internal Determinants of Profitability

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Constant	0.00224*** (0.000530)	0.000890 (0.000718)	-0.00274*** (0.000222)	0.0161*** (0.000791)	0.0192*** (0.000815)	0.0274*** (0.000933)	0.00996*** (0.00112)	0.0141*** (0.00122)	0.00258 (0.00167)	0.00919*** (0.00106)
NIM _{t-1}	0.807*** (0.0102)	0.868*** (0.0257)	0.746*** (0.00367)	0.732*** (0.00550)	0.721*** (0.00526)	0.720*** (0.00370)	0.693*** (0.00405)	0.693*** (0.00400)	0.684*** (0.00391)	0.686*** (0.00447)
CAP	0.0315*** (0.00856)	-0.0126 (0.0186)	0.0708*** (0.00352)	0.0606*** (0.00332)	0.0795*** (0.00306)	0.0442*** (0.00263)	0.0464*** (0.00347)	0.0167*** (0.00398)	0.0464*** (0.00353)	0.0536*** (0.00356)
CAP ²	-0.0343* (0.0203)	0.0891 (0.0684)	-0.118*** (0.00856)	-0.142*** (0.00777)	-0.194*** (0.00690)	-0.162*** (0.00635)	-0.0868*** (0.0114)	-0.00502 (0.0118)	-0.0744*** (0.0113)	-0.0985*** (0.0109)
Critical Value CAP	0.4592	-	0.3000	0.2134	0.2049	0.1364	0.2673	-	0.3118	0.2721
OWN1		0.0106*** (0.00359)								
OWN1 ²		-0.00565 (0.00449)								
INSOWN			0.0277*** (0.000495)	0.0183*** (0.000525)	0.0177*** (0.000573)	0.0156*** (0.000559)	0.0159*** (0.000563)	0.0184*** (0.000532)	0.0160*** (0.000585)	0.0156*** (0.000604)
INSOWN ²			-0.0284*** (0.000556)	-0.0187*** (0.000589)	-0.0175*** (0.000627)	-0.0138*** (0.000586)	-0.0138*** (0.000592)	-0.0161*** (0.000542)	-0.0135*** (0.000619)	-0.0135*** (0.000644)
Critical Value INSOWN			0.4877	0.4893	0.5057	0.5652	0.5761	0.5714	0.5926	0.5778
LnTA				-0.000962*** (4.04e-05)	-0.000995*** (3.98e-05)	-0.000974*** (3.25e-05)	-0.000392*** (5.53e-05)	-0.000488*** (5.63e-05)	-0.000374*** (5.54e-05)	-0.000229*** (6.42e-05)
LTA					-0.00494*** (0.000421)	-0.00323*** (0.000454)	-0.000377 (0.000668)	0.00181*** (0.000695)	0.00627*** (0.00137)	-0.00150** (0.000755)
DEPTA						-0.00899*** (0.000696)	-0.00438*** (0.000858)	-0.00183** (0.000817)	-0.00484*** (0.000897)	-0.00490*** (0.000898)
CreRisk							0.305*** (0.00654)	0.357*** (0.00801)	0.306*** (0.00697)	0.308*** (0.00680)
Divers1								-0.0104*** (0.000944)		
Divers2									0.00476*** (0.000695)	
Divers3										-0.00207*** (0.000265)
Observations	2,643	1,397	2,283	2,283	2,262	2,249	2,238	2,234	2,238	2,234
Number of iden	260	243	248	248	238	236	235	235	235	235
Temporal Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No Instruments	80	75	159	159	159	159	159	159	159	159
Wald	61167	3321.04	64042.39	99489.71	137472.01	12236.54	65151.97	60775.48	55052.05	61166.99
p-value Wald	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AR(1)	-5.51	-4.55	-5.45	-5.48	-5.44	-5.49	-5.46	-5.46	-5.47	-5.401
p-value AR(1)	0.000	0.000	0.000	0.000	0.008	0.000	0.000	0.000	0.000	0.000
AR(2)	-1.80	-2.000	-2.06	-2.20	-2.23	-2.35	-2.769	-2.88	-2.84	-2.769
p-value AR(2)	0.562	0.460	0.390	0.280	0.262	0.197	0.562	0.410	0.408	0.156
Sargan	493.10	256.00	583.74	577.18	580.15	575.71	494.25	449.91	494.47	493.1
p-value Sargan	0.000	0.00	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Hansen	187.8	127.85	196.26	200.02	199.46	202.38	188.13	181.56	183.05	187.8
p-value Hansen	0.150	0.150	0.174	0.252	0.204	0.301	0.170	0.319	0.270	0.150
Lind-Mehlum CAP	0.28	-	22.63	19.25	25.66	15.84	4.75	-	3.42	5.61
p-value Lind-Mehlum	0.391	-	0.000	0.000	0.000	0.000	0.000	-	0.000	0.000
Lind-Mehlum	-	0.13	41.71	25.84	23.28	17.57	16.76	22.57	14.88	15.14
INSOWN										
p-value Lind-Mehlum	-	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 4. Regression Analysis for Internal and External Determinants of Profitability

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
		Development of the Financial System						Legal Enforcement and Regulatory System					
		Banking System			Capital Markets								
Constant	0.0108*** (0.00118)	0.00788*** (0.00111)	0.00913*** (0.00113)	0.00781*** (0.00103)	0.00917*** (0.00107)	0.00874*** (0.00102)	0.00809*** (0.00106)	0.0141*** (0.00106)	0.00325*** (0.00101)	0.0106*** (0.00102)	0.0106*** (0.00103)	0.0144*** (0.00103)	0.0119*** (0.000942)
NIM _{t-1}	0.687*** (0.00449)	0.639*** (0.00451)	0.681*** (0.00504)	0.634*** (0.00478)	0.685*** (0.00448)	0.686*** (0.00460)	0.676*** (0.00506)	0.611*** (0.00508)	0.547*** (0.00488)	0.615*** (0.00486)	0.619*** (0.00463)	0.555*** (0.00520)	0.587*** (0.00516)
CAP	0.0573*** (0.00402)	0.0428*** (0.00389)	0.0523*** (0.00377)	0.0426*** (0.00372)	0.0535*** (0.00363)	0.0499*** (0.00357)	0.0518*** (0.00368)	0.0408*** (0.00441)	0.0431*** (0.00324)	0.0468*** (0.00398)	0.0413*** (0.00361)	0.0389*** (0.00428)	0.0387*** (0.00400)
CAP ²	-0.106*** (0.0110)	-0.0654*** (0.0108)	-0.0956*** (0.0109)	-0.0664*** (0.0104)	-0.0968*** (0.0112)	-0.0868*** (0.0107)	-0.0903*** (0.0110)	-0.0563*** (0.0122)	-0.0480*** (0.00936)	-0.0741*** (0.0109)	-0.0554*** (0.0101)	-0.0500*** (0.0118)	-0.0593*** (0.0109)
Critical Value CAP	0.2703	0.3272	0.2735	0.3208	0.2763	0.2874	0.2868	0.3623	0.4490	0.3158	0.3727	0.3890	0.3263
INSOWN	0.0152*** (0.000619)	0.0138*** (0.000562)	0.0157*** (0.000607)	0.0157*** (0.000611)	0.0152*** (0.000669)	0.0156*** (0.000622)	0.0153*** (0.000612)	0.0146*** (0.000563)	0.0128*** (0.000522)	0.0139*** (0.000601)	0.0137*** (0.000568)	0.0128*** (0.000569)	0.0102*** (0.000557)
INSOWN ²	-0.0134*** (0.000670)	-0.0137*** (0.000637)	-0.0138*** (0.000652)	-0.0157*** (0.000685)	-0.0132*** (0.000711)	-0.0134*** (0.000659)	-0.0138*** (0.000672)	-0.0132*** (0.000618)	-0.0135*** (0.000625)	-0.0142*** (0.000703)	-0.0142*** (0.000647)	-0.0148*** (0.000678)	-0.0117*** (0.000633)
Critical Value INSOWN	0.5672	0.5036	0.5688	0.5000	0.5758	0.5821	0.5543	0.5530	0.4741	0.4894	0.4824	0.4324	0.4359
LnTA	-0.000269*** (6.52e-05)	3.31e-05 (7.24e-05)	-0.000214*** (6.57e-05)	3.71e-05 (6.25e-05)	-0.000219*** (6.38e-05)	-0.000167*** (6.15e-05)	-0.000133*** (6.21e-05)	-0.000495*** (5.55e-05)	7.05e-05 (4.95e-05)	-0.000178*** (5.38e-05)	-0.000206*** (5.58e-05)	-0.000283*** (4.85e-05)	-0.000272*** (5.65e-05)
LTA	-0.00155* (0.000803)	0.000472 (0.000836)	-0.00150* (0.000812)	3.19e-05 (0.000808)	-0.00110 (0.000777)	-0.00163** (0.000749)	-0.000889 (0.000764)	0.000295 (0.000785)	0.00485*** (0.000809)	0.00109 (0.000781)	0.00136* (0.000719)	0.00135** (0.000675)	0.00209*** (0.000804)
DEPTA	-0.00500*** (0.000931)	-0.00289*** (0.00101)	-0.00459*** (0.000937)	-0.00289*** (0.000985)	-0.00528*** (0.000909)	-0.00468*** (0.000913)	-0.00467*** (0.000948)	-0.00211*** (0.000741)	-0.00144** (0.000724)	-0.00416*** (0.000892)	-0.00350*** (0.000860)	-0.00344*** (0.000823)	-0.00278*** (0.000859)
CredRisk	0.319*** (0.00710)	0.298*** (0.00801)	0.305*** (0.00731)	0.304*** (0.00836)	0.301*** (0.00715)	0.305*** (0.00681)	0.309*** (0.00772)	0.278*** (0.00695)	0.280*** (0.00859)	0.282*** (0.00774)	0.276*** (0.00861)	0.273*** (0.00823)	0.261*** (0.00778)
Divers3	-0.00203*** (0.000277)	-0.00219*** (0.000270)	-0.00218*** (0.000274)	-0.00272*** (0.000287)	-0.00210*** (0.000272)	-0.00204*** (0.000289)	-0.00218*** (0.000282)	-0.00178*** (0.000264)	-0.00365*** (0.000268)	-0.00102*** (0.000262)	-0.00117*** (0.000283)	-0.00122*** (0.000238)	-0.00113*** (0.000261)
BankConc	-2.30e-05*** (6.14e-06)												
DMBAGDP		-3.10e-05*** (2.75e-06)											
OFAGDP			-1.40e-06* (8.30e-07)										
PCRDBoFGDP				-2.84e-05*** (2.20e-06)									
STMKTcap					4.72e-07 (1.96e-06)								
STTO						-5.23e-06*** (1.49e-06)							
PRBOND							-2.46e-05*** (7.60e-06)						
VA								-0.00250*** (0.000121)					
PS									-0.00399*** (0.000117)				
GE										-0.00282*** (0.000154)			
RQ											-0.00376*** (0.000184)		
RL												-0.00419*** (0.000127)	
CC													-0.00312***

(0.000146)

Observations	2,234	2,234	2,234	2,234	2,234	2,234	2,234	2,234	2,234	2,234	2,234	2,234	2,234
Number of iden	235	235	235	235	235	235	235	235	235	235	235	235	235
Temporal Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No Instruments	159	159	159	159	159	159	159	159	159	159	159	159	159
Wald-test	64156.51	69402.71	61346.68	56022.02	56340	63422.81	58701.7	55271.44	78906.22	80572.20	79664.58	67456.59	56340.00
p-value Wald-test	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AR(1)	-5.41	-5.49	-5.40	-5.34	-5.38	-5.36	-5.40	-5.37	-5.56	-5.44	-5.47	-5.37	-5.31
p-value AR(1)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AR(2)	-2.81	-2.52	-2.75	-2.68	-2.75	-2.78	-2.78	-2.82	-2.43	-2.70	-2.70	-2.56	-2.589
p-value AR(2)	0.162	0.108	0.206	0.170	0.362	0.155	0.162	0.150	0.154	0.147	0.177	0.192	0.195
Sargan-test	489.23	460.39	495.87	512.83	494.58	494.09	497.31	518.44	496.03	517.73	494.31	526.26	522.55
p-value Sargan-test	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Hansen-test	186.61	183.93	187.76	188.36	185.42	18.40	187.34	187.97	187.41	187.68	19014	192.09	190.40
p-value Hansen-test	0.152	0.210	0.130	0.120	0.174	0.121	0.143	0.130	0.141	0.130	0.110	0.155	0.130
Lind-Mehlum CAP	6.15	7.51	5.42	3.32	5.25	4.80	4.84	1.92	0.85	3.69	2.09	1.49	2.80
p-value Lind-Mehlum	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.0276	0.199	0.000	0.018	0.069	0.003
Lind-Mehlum INSOWN	14.64	19.28	15.57	18.87	13.59	14.62	15.17	15.68	17.97	16.28	18.31	19.59	16.81
p-value Lind-Mehlum	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

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Table 5. Principal Component Factoring Analysis

Panel A: Development of the Financial System					
Variables	Factor	Eigenvalue	Difference	Proportion	Cumulative
DMBAGDP	Factor1	3.790	2.837	0.632	0.632
OFAGDP	Factor2	0.952	0.134	0.159	0.790
PCRDBOFGDP	Factor3	0.818	0.563	0.136	0.927
STMKCAP	Factor4	0.255	0.079	0.042	0.969
STTO	Factor5	0.175	0.165	0.029	0.998
PRBOND	Factor6	0.010	0.000	0.002	1.000

Panel B: Legal and Regulatory System					
VA	Factor1	5.246	4.696	0.874	0.874
PS	Factor2	0.550	0.455	0.092	0.966
GE	Factor3	0.095	0.036	0.016	0.982
RQ	Factor4	0.059	0.026	0.010	0.992
RL	Factor5	0.032	0.015	0.005	0.997
CC	Factor6	0.018	0.000	0.003	1.000

Table 6. Regression Analysis Including Factor

VARIABLES	(1)	(2)	(3)
Constant	0.00536*** (0.00104)	0.00966*** (0.000965)	0.0199*** (0.00127)
NIM _{t-1}	0.654*** (0.00474)	0.563*** (0.00520)	0.569*** (0.00613)
CAP	0.0446*** (0.00377)	0.0386*** (0.00390)	0.0544*** (0.00383)
CAP ²	-0.0711*** (0.0107)	-0.0465*** (0.0108)	-0.0844*** (0.0107)
<i>Critical Value CAP</i>	<i>0.3136</i>	<i>0.4151</i>	<i>0.3223</i>
INSOWN	0.0153*** (0.000612)	0.0124*** (0.000584)	0.0112*** (0.000510)
INSOWN ²	-0.0146*** (0.000689)	-0.0138*** (0.000683)	-0.0110*** (0.000594)
<i>Critical Value INSOWN</i>	<i>0.5240</i>	<i>0.4493</i>	<i>0.5091</i>
LnTA	0.000107 (5.89e-05)	-0.000263*** (4.79e-05)	-0.000877*** (6.76e-05)
LTA	-0.000509 (0.000779)	0.00246*** (0.000725)	0.00221*** (0.000801)
Depta	-0.00372*** (0.000987)	-0.00190** (0.000747)	-0.00360*** (0.000767)
CredRisk	0.294*** (0.00812)	0.265*** (0.00865)	0.275*** (0.00919)
Divers3	-0.00229*** (0.000280)	-0.00130*** (0.000246)	-0.000378 (0.000280)
FinDevFact1	-0.00125*** (0.000123)		-0.00335*** (0.000229)
LegalSysFact1		-0.00334*** (0.000116)	-0.00538*** (0.000201)
Observations	2,234	2,234	2,234
Number of iden	235	235	235
Temporal Effect	Yes	Yes	Yes
Country Effect	Yes	Yes	Yes
No Instruments	159	159	159
Wald-test	61850.10	64819.04	67354.37
p-value Wald-test	0.000	0.000	0.000
AR(1)	-5.33	-5.42	-5.576
p-value AR(1)	0.000	0.000	0.000
AR(2)	-2.73	-2.643	-2.643
p-value AR(2)	0.223	0.142	0.223
Sargan-test	511.34	522.29	488.75
p-value Sargan-test	0.000	0.000	0.000
Hansen-test	188.55	190.56	188.22
p-value Hansen-test	0.106	0.144	0.116
Lind-Mehlum CAP	3.54	1.17	4.03
p-value Lind-Mehlum	0.000	0.121	0.000
Lind-Mehlum INSOWN	16.75	17.90	14.40
p-value Lind-Mehlum	0.000	0.000	0.000

Table 7: Comparative Analysis Based on the Regulatory Systems

Panel A				
Countries	Obs.	Mean		
CHN	174	-0.540		
IDN	333	-0.560		
IND	389	-0.275		
JPN	1,071	1.175		
KOR	102	0.721		
LKA	104	-0.344		
MYS	129	0.359		
PHL	168	-0.449		
SGP	78	1.489		
THA	131	-0.148		
TWN	232	0.884		
Total	2911	0.406		

Panel B				
VARIABLES	(1)	(2)	(3)	(4)
	High Standards	Low Standards	High Standards	Low Standards
Constant	-0.00618*** (0.000378)	0.0239*** (0.00265)	-0.00854*** (0.000508)	0.00161 (0.00249)
NIM _{t-1}	0.696*** (0.00245)	0.578*** (0.0141)	0.684*** (0.00271)	0.537*** (0.0125)
CAP	0.0104*** (0.000913)	0.226*** (0.0110)	0.0181*** (0.000692)	0.210*** (0.00967)
CAP ²	0.0291*** (0.00153)	-0.750*** (0.0495)	0.0212*** (0.00206)	-0.691*** (0.0452)
Critical Value CAP	-	0.1507	-	0.1520
INSOWN	0.00956*** (0.000148)	-0.000649 (0.00197)	0.00953*** (0.000157)	-0.00677*** (0.00211)
INSOWN ²	-0.00992** (0.000181)	-0.00222 (0.00222)	-0.0110*** (0.000194)	0.00336* (0.00203)
Critical Value INSOWN	0.4819	-	0.4332	-
LnTA	0.000137*** (1.73e-05)	-0.00107*** (8.65e-05)	0.000284*** (2.65e-05)	-0.000564*** (9.75e-05)
LTA	0.00411** (0.000119)	0.00304*** (0.000761)	0.00435** (0.000153)	0.00946*** (0.000784)
Depta	-0.00358* (0.000114)	-0.00946** (0.00220)	-0.00505*** (0.000128)	-0.0136*** (0.00189)
CredRisk	0.156*** (0.00138)	0.437*** (0.0146)	0.135*** (0.00135)	0.442*** (0.0133)
Divers3	-0.000183*** (5.05e-05)	-0.00416* (0.000593)	-0.000943*** (6.35e-05)	-0.00321*** (0.000581)
FinDevFact1			-0.000258** (2.96e-05)	-0.00181*** (0.000640)
LegalSysFact1			-0.00169*** (4.01e-05)	-0.0132*** (0.000737)
Observations	1,291	943	1,291	943
Number of iden	121	114	121	114
Temporal Effect	Yes	Yes	Yes	Yes
Country Effect	Yes	Yes	Yes	Yes
No Instruments	159	159	159	159
Wald-test	931520.61	763526.54	406243	52246.26
p-value Wald-test	0.000	0.000	0.000	0.000
AR(1)	-2.69	-4.919	-2.69	-4.18
p-value AR(1)	0.000	0.000	0.007	0.000
AR(2)	-0.40	-1.44	-0.30	-2.334
p-value AR(2)	0.691	0.196	0.765	0.019
Sargan-test	275.56	291.12	271.78	268.4
p-value Sargan-test	0.000	0.000	0.000	0.000
Hansen-test	118.04	93.32	118.13	93.32
p-value Hansen-test	0.967	0.921	0.956	0.855
Lind-Mehlum CAP	EOI	12.94	EOI	13.16
p-value Lind-Mehlum	-	0.000	-	0.000
Lind-Mehlum INSOWN	45.74	EOI	51.90	EOI
p-value Lind-Mehlum	0.00	-	0.000	-

EOI: Extremum outside interval - trivial rejection of H0.

Table 8. Robustness Checks with ROA

VARIABLES	(1)	(2)	(3)
Constant	0.00197* (0.00101)	0.00622*** (0.000867)	0.00574*** (0.000974)
ROA _{t-1}	0.335*** (0.00765)	0.301*** (0.00820)	0.301*** (0.00817)
CAP	0.100*** (0.00357)	0.108*** (0.00374)	0.107*** (0.00381)
CAP ²	-0.212*** (0.00939)	-0.223*** (0.00998)	-0.223*** (0.0100)
<i>Critical Value CAP</i>	<i>0.2358</i>	<i>0.2422</i>	<i>0.2399</i>
INSOWN	0.00397*** (0.000628)	0.00734*** (0.000645)	0.00719*** (0.000655)
INSOWN ²	-0.00450** (0.000670)	-0.00782*** (0.000681)	-0.00764*** (0.000698)
<i>Critical Value INSOWN</i>	<i>0.4411</i>	<i>0.4693</i>	<i>0.4705</i>
LnTA	0.000414*** (6.64e-05)	8.54e-05 (5.31e-05)	0.000132** (6.64e-05)
LTA	-0.0127* (0.000506)	-0.0121*** (0.000596)	-0.0122*** (0.000607)
Depta	-0.00357*** (0.000759)	-0.00277*** (0.000844)	-0.00300*** (0.000857)
CredRisk	-0.0940* (0.00678)	-0.101** (0.00619)	-0.102** (0.00621)
Divers3	0.00468 (0.000226)	0.00580 (0.000246)	0.00574 (0.000251)
FinDevFact1	-0.00206** (0.000134)		-0.000205 (0.000197)
LegalSysFact1		-0.00237*** (0.000149)	-0.00222*** (0.000204)
Observations	2,392	2,392	2,392
Number of iden	237	237	237
Temporal Effect	Yes	Yes	Yes
Country Effect	Yes	Yes	Yes
No Instruments	161	161	161
Wald-test	10114	10114	10114
p-value Wald-test	0	0	0
AR(1)	-6.558	-6.558	-6.558
p-value AR(1)	5.44e-11	5.44e-11	5.44e-11
AR(2)	0.114	0.114	0.114
p-value AR(2)	0.909	0.909	0.909
Sargan-test	421.2	421.2	421.2
p-value Sargan-test	0	0	0
Hansen-test	185.6	185.6	185.6
p-value Hansen-test	0.0196	0.0196	0.0196
Lind-Mehlum CAP	17.03	16.16	16.15
p-value Lind-Mehlum	0.000	0.000	0.000
Lind-Mehlum INSOWN	6.33	10.93	10.30
p-value Lind-Mehlum	0.00	0.000	0.000

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